

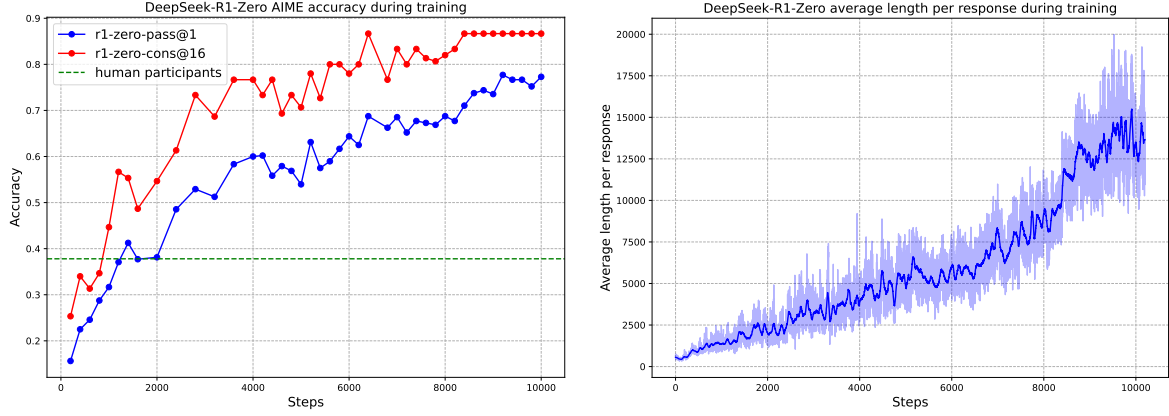


Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

responses against a suite of predefined test cases, thereby generating objective feedback on correctness.

Format rewards complement the accuracy reward model by enforcing specific formatting requirements. In particular, the model is incentivized to encapsulate its reasoning process within designated tags, specifically ‘<think>’ and ‘</think>’. This ensures that the model’s thought process is explicitly delineated, enhancing interpretability and facilitating subsequent analysis.

$$Reward_{rule} = Reward_{acc} + Reward_{format} \quad (4)$$

The accuracy, reward and format reward are combined with the same weight. Notably, we abstain from applying neural reward models—whether outcome-based or process-based—to reasoning tasks. This decision is predicated on our observation that neural reward models are susceptible to reward hacking during large-scale reinforcement learning. Moreover, retraining such models necessitates substantial computational resources and introduces additional complexity into the training pipeline, thereby complicating the overall optimization process.

2.3. Incentivize Reasoning Capability in LLMs

Specifically, we apply the RL technique on the DeepSeek-V3 base to train DeepSeek-R1-Zero. During training, we design a straightforward template, to require DeepSeek-R1-Zero to first produce a reasoning process, followed by the final answer. We intentionally limit our constraints to this structural format, avoiding any content-specific biases to ensure that we can accurately observe the model’s natural progression during the RL process.

Figure 1(a) depicts the performance trajectory of DeepSeek-R1-Zero on the AIME 2024 benchmark throughout the RL training process, where the average pass@1 score on AIME 2024 shows a significant increase, jumping from an initial 15.6% to 77.9%. In addition, by leveraging the self-consistency decoding (Wang et al., 2023c), the model’s performance can be

Table 2 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think>
To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let’s start by squaring both ...
$(\sqrt{a - \sqrt{a + x}})^2 = x^2 \implies a - \sqrt{a + x} = x^2$.
Rearrange to isolate the inner square root term:
$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$
...
Wait, wait. Wait. That’s an aha moment I can flag here.
Let’s reevaluate this step-by-step to identify if the correct sum can be ...
We started with the equation:
$\sqrt{a - \sqrt{a + x}} = x$
First, let’s square both sides:
$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$
Next, I could square both sides again, treating the equation: ...
...

further improved, achieving an accuracy of 86.7%. This performance significantly surpasses the average performance across all human competitors. Besides the math competitions, as shown in Figure 10, DeepSeek-R1-Zero also achieves remarkable performance in coding competitions and graduate-level biology, physics, and chemistry problems. These results underscore the effectiveness of RL in enhancing the reasoning capabilities of large language models.

The self-evolution of DeepSeek-R1-Zero exemplifies how RL can autonomously enhance a model’s reasoning capabilities.

As shown in Figure 1(b), DeepSeek-R1-Zero exhibits a steady increase in thinking time throughout training, driven solely by intrinsic adaptation rather than external modifications. Leveraging long CoT, the model progressively refines its reasoning, generating hundreds to thousands of tokens to explore and improve its problem-solving strategies.

The increase in thinking time fosters the autonomous development of sophisticated behaviors. Specifically, DeepSeek-R1-Zero increasingly exhibits advanced reasoning strategies such as reflective reasoning and systematic exploration of alternative solutions (see Figure 9(a) in Supplementary C.2 for details), significantly boosting its performance on verifiable tasks like math and coding. Notably, during training, DeepSeek-R1-Zero exhibits an “aha moment” (Table 2), characterized by a sudden increase in the use of the word “wait” during reflections (see Figure 9(b) in Supplementary C.2 for details). This moment marks a distinct change in reasoning patterns and clearly shows the self-evolution process of DeepSeek-R1-Zero.

The self-evolution of DeepSeek-R1-Zero underscores the power and beauty of RL: rather than explicitly teaching the model how to solve a problem, we simply provide it with the right incentives, and it autonomously develops advanced problem-solving strategies. This serves as a reminder of the potential of RL to unlock higher levels of capabilities in LLMs, paving the way for more autonomous and adaptive models in the future.

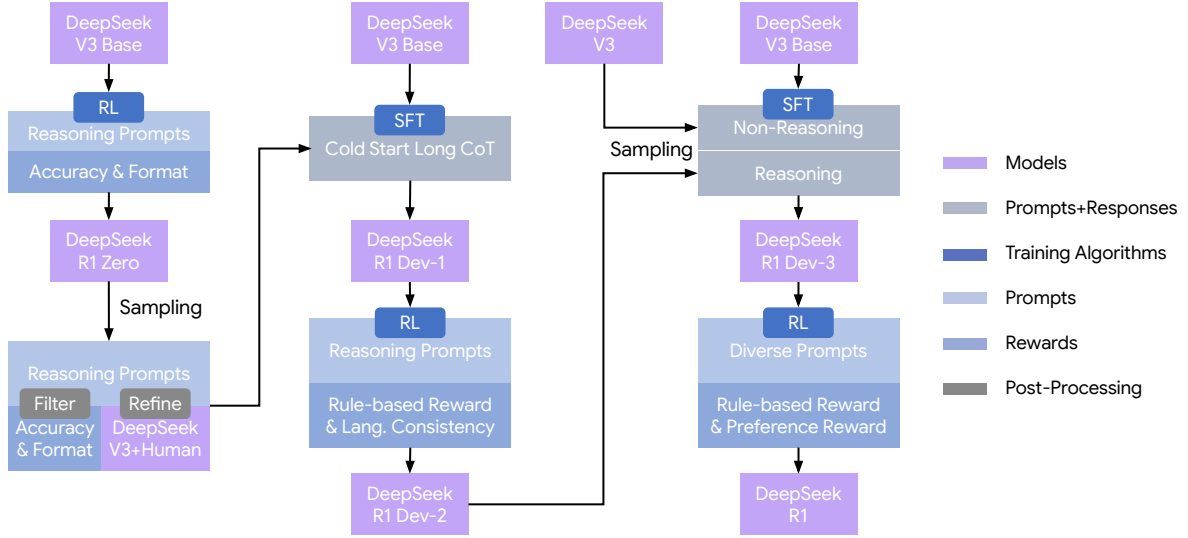


Figure 2 | The multi-stage pipeline of DeepSeek-R1. A detailed background on DeepSeek-V3 Base and DeepSeek-V3 is provided in Supplementary A.1. The models DeepSeek-R1 Dev1, Dev2, and Dev3 represent intermediate checkpoints within this pipeline.

3. DeepSeek-R1

Although DeepSeek-R1-Zero exhibits strong reasoning capabilities, it faces several issues. DeepSeek-R1-Zero struggles with challenges like poor readability, and language mixing, as DeepSeek-V3-Base is trained on multiple languages, especially English and Chinese. To address these issues, we develop DeepSeek-R1, whose pipeline is illustrated in Figure 2.

In the initial stage, we collect thousands of cold-start data that exhibits a conversational, human-aligned thinking process. RL training is then applied to improve the model performance with the conversational thinking process and language consistency. Subsequently, we apply rejection sampling and SFT once more. This stage incorporates both reasoning and non-reasoning datasets into the SFT process, enabling the model to not only excel in reasoning tasks but also demonstrate advanced writing capabilities. To further align the model with human preferences, we implement a secondary RL stage designed to enhance the model’s helpfulness and harmlessness while simultaneously refining its reasoning capabilities.

The remainder of this section details the key components of this pipeline: Section 3.1 introduces the Reward Model utilized in our RL stages, and Section 3.2 elaborates on the specific training methodologies and implementation details. Data we used in this stage is detailed in Supplementary B.3.

3.1. Model-based Rewards

For general data, we resort to reward models to capture human preferences in complex and nuanced scenarios. We build upon the DeepSeek-V3 pipeline and adopt a similar distribution of preference pairs and training prompts. For helpfulness, we focus exclusively on the final summary, ensuring that the assessment emphasizes the utility and relevance of the response to the user while minimizing interference with the underlying reasoning process. For harmlessness, we evaluate the entire response of the model, including both the reasoning process and the summary, to identify and mitigate any potential risks, biases, or harmful content that may arise